



Object Detection Algorithms & Techniques

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Abstract

Object detection is a computer vision technology that allows us to determine the type or location of objects in images or videos, face recognition, object recognition, driverless vehicles, security systems, etc. It has various applications such as. This article aims to examine the ways of detecting objects and to provide an overview of traditional, two-stage, and single-stage object detection techniques. In addition, it aims to make a simple comparison by giving information about the most common applications in object detection applications.

Nesne Algılama Algoritmaları ve Teknikleri

Anahtar Kelimeler

Nesne algılama
Nesne sınıflandırması
Derin Öğrenme
Evrşimli sinir ağları
CNN

Öz

Nesne algılama, görüntülerdeki veya videolardaki nesnelerin türünü veya konumlarını belirlememizi sağlayan bir bilgisayarlı görme teknolojisidir, yüz tanıma, nesne tanıma, sürücüsüz araçlar, güvenlik sistemleri vb. gibi çeşitli uygulamalara sahiptir. Bu makalede, nesneleri keşfetme yollarının incelenmesi, geleneksel, iki aşamalı ve tek aşamalı nesneleri algılama tekniklerine genel bir bakış sağlanması amaçlanmıştır. Ayrıca nesne algılama uygulamalarında en yaygın olan uygulamalar hakkında bilgi vererek basit bir karşılaştırma yapmayı hedeflenmiştir.

Introduction

Object detection is one of the computer vision tasks where that attempts to discover a specific object in a visual image or video (K. Li & CAO, 2020). The goal of object detection is to develop a computational model or technique that provides one of the simplest pieces of information needed by computer vision applications, which is one of the fundamental problems of computer vision applications., What is there? And where is it located? (Z. Zou, Shi, Guo, & Ye, 2019). Detecting objects is one of the best-known challenges in computer vision (Krizhevsky, Sutskever, & Hinton, 2017). Over the last decade, object discovery has become the most active and growing area of research in computer vision (Diwakar & Raj, 2022). Therefore, object detection, segmentation, object tracking, image annotation, event detection, event recognition, etc. It forms the basis for solving complex or high-level computer vision tasks (L. Liu et al., 2020). In terms of application, object exploration can be divided into two parts: First, "general object exploration", which aims to explore different types of objects within a unified framework to simulate human vision and perception in such exploration, second is "perception applications", such exploration techniques are also pedestrian detection, face detection, text detection, etc. aims to discover predefined types (X. Zou, 2019). It supports a wide range of applications, including object discovery, robotic vision, autonomous driving, human-computer interaction, intelligent video monitoring and medical image sensing (Xiao et al., 2020). Object detection techniques are extensive, and there are many algorithms proposed for computer vision that are being rapidly developed. Testing techniques takes time and effort to choose the most suitable technique for object detection. Since each technology has advantages and disadvantages, there are many factors that affect the selection of the appropriate technology (Noman, Stankovic, & Tawfik, 2019). This article aims to provide some techniques based on both traditional object detection methods and deep learning of object detection. Firstly, traditional techniques are presented in the first part. In the second part, deep learning-based techniques including two-stage and one-stage methods are presented, in the Third Part a table (simple comparison) of the advantages and disadvantages of the most common techniques is presented. In the fourth chapter, the most important applications for discovering items are also mentioned. Finally, references are added.

Object Detection Algorithms

It has been unanimously agreed that in the last two decades, developments in the field of object discovery have been made in two historical periods, one is the era of traditional object discovery before 2014, and the other is the era of deep learning-based object discovery after 2014)Z. Zou et al., 2019(.

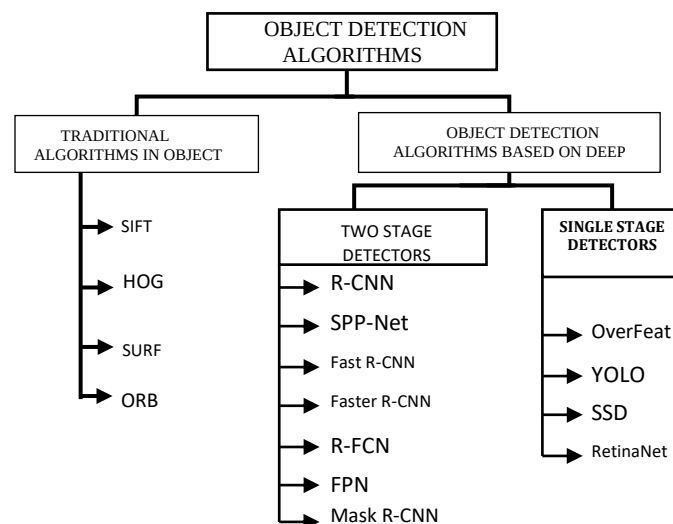


Figure 1. Object Detection Algorithms

Traditional Algorithms in Object Detection

Most of the traditional object detection algorithms are developed using hand-crafted features; Due to the lack of advanced techniques to represent images before 2012. Therefore, the only option for researchers was to design vector representations of highly complex features along with a wide range of acceleration techniques to compensate for the mismatch introduced due to the availability of limited computational resources (Sharma &

Mir, 2020). There are three stages in traditional object discovery methods, and these stages are explained as follows (MLK, 2020; Zhao, Zheng, Xu, & Wu, 2019):

1) Determining the area: Since the size and aspect ratio of objects are different, different areas may occur in the image. Therefore, at the first stage, it is necessary to determine the area of the object; As a result, the entire image is scanned using a multi-scale sliding window approach to detect objects. However, this approach has a high computational cost and also results in a large number of unnecessary options.

2) Extracting features: Once the object is found, feature extraction is performed to provide a robust representation. Methods such as HOG to extract object recognition features (Dalal & Triggs, 2005) and SIFT (Lowe, 1999) are used; To provide meaningful representation. However, due to contrasting backgrounds, lighting environment, and perspective contrasts, it is very difficult to manually create a comprehensive feature and descriptor that accurately identifies all types of objects.

3) Classification: In the classification phase, the classifier is used as an example to identify target objects and create models in a more orderly and meaningful way for visual perception.

A. SIFT

This method was first introduced in 1999 by Canadian Professor David G. Lowe. SIFT can find fixed principal points that do not vary with position, size, and rotation. And it's powerful for documenting transitions (changes in size, rotation, cropping, and position) and changes in lighting. Therefore, this technique can be used for object (Lowe, 1999).

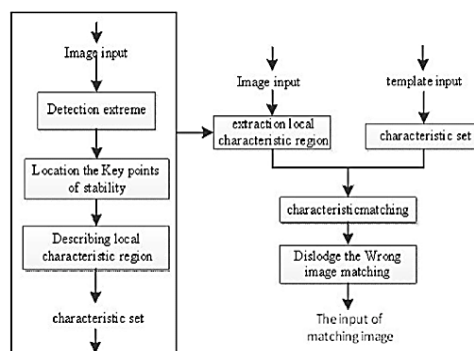


Figure 2. SIFT algorithm flow chart.

B. HOG

It was first introduced in 2005 by N. Dalal and B. Triggs. It is primarily designed to perform calculations on a dense grid of evenly spaced cells. HOG can be used to detect various classes of objects, but pedestrian detection has been the primary driving force behind the development of HOG technology. The best feature of the HOG detector is that it can be used to detect objects of different sizes, and this can be achieved by rescaling the input image several times without changing the size of the detection window (Dalal & Triggs, 2005; Sharma & Mir, 2020)

C. SURF

It was first introduced by Herbert Bay and others. In 2006. SURF is a fast and powerful algorithm for static local representation of similarity and image comparison. The main interest of the SURF approach lies in the fast calculation of operators using box filters, thus enabling real-time applications such as tracking and object recognition. It is a much faster technology than SIFT (Bay, Tuytelaars, & Gool, 2006; Di Calafiori et al., 2009).

D. ORB

In 2012, Li Xiaohong and others proposed a new method for object detection under dynamic scenes based on ORB. It uses the eight-parameter rotation model, combines it with the least squares method to solve for the overall motion parameters for motion compensation, and achieves the final motion goal using the frame difference method. Experiments show that this method not only incorporates SURF but also increases detection levels and real-time efficiency. It can also quickly and reliably track moving objects in real time (K. Li & Cao, 2020; X. Li, Xie, Jia, & Zhang, 2013).

Object Detection Algorithms Based on Deep Learning

Undoubtedly, traditional ways of discovering things still work effectively, but deep learning in the field of computer vision has given a huge boost to development and is now a part of our daily lives. Neural-based methods have added new features, increased accuracy and efficiency, and opened many ways to develop new technologies

using deep learning methods such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) (Diwakar & Raj, 2022).

(Object discovery devices) that are based on deep learning can be divided into two main categories: two-stage sensor device and single-stage sensor device. Speaking of the two-stage sensor device, in the first stage a small set of basis propositions is generated, in the next stage feature vectors are extracted from these propositions, then encoded using deep convolutional neural networks, and then predefined class predictions are made. On the other hand, a single-stage sensor device usually considers all locations in the input image as potential target objects and attempts to classify each area of interest as a target object or just a background. Single-stage sensor devices are much faster and well-suited for real-time object detection applications, but generally perform relatively poorly compared to two-stage sensor devices, which provide impressive results on many publicly available data sets (Sharma & Mir, 2020).

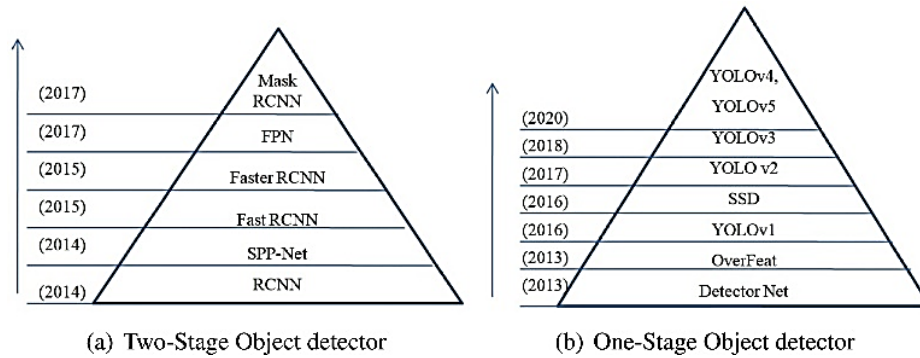


Figure 3. Classification of general object detection models (Kaur & Singh, 2022).

A. Two-stage detectors

1) R-CNN

First presented in 2014 by Girshick et al. R-CNN stands for Region Based Convolutional Neural Network. Object detection consists of two separate tasks: classification and localization. First, it identifies a controllable number of candidates for the bounding box area ("region of interest" or "ROI") using a selective search. Then, it independently extracts CNN features from each region for classification and uses the region recommendations to find objects within the image (Girshick, Donahue, Darrell, & Malik, 2014).

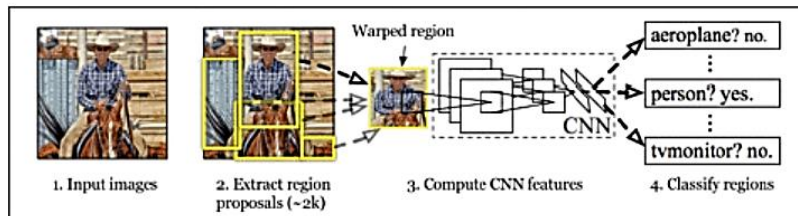


Figure 4. Architecture of R-CNN (Girshick et al., 2014)

2) SPP-Net

First presented in 2014 by He and others.

SPP-Net is a bypass neural architecture that uses spatial hierarchy clustering to remove the fixed size constraint of the network. Specifically, we add an SPP layer on top of the last convolution layer. The SPP layer collects features and produces a fixed-length output that is then fed into fully connected layers. In other words, we perform some information gathering at a deeper stage of the network hierarchy (between convolutional layers and fully connected layers) to avoid the need for clipping or convolution at the beginning (He, Zhang, Ren, & Sun, 2015; Kleban, Xie, & Ma, 2008).

3) Fast R-CNN

Fast R-CNN is an object detector developed specifically in 2015 by Ross Girshick, an AI researcher at Facebook and a former researcher at Microsoft. Fast R-CNN overcomes many problems with R-CNN. As the name suggests, one of the advantages of Fast R-CNN over R-CNN is its speed (Girshick, 2015). In a Fast R-CNN architecture, as shown in Fig. 3, whole image is given as input along with object proposals. Several convolutional and max-pooling layers are used to process the image and produce conv feature map as output.

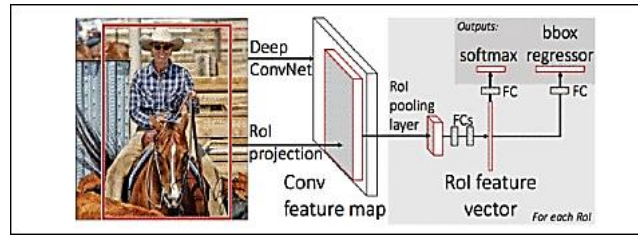


Figure 5. Architecture of Fast R-CNN(Girshick, 2015)

4) Faster R-CNN

Shortly after introducing Fast R-CNN technology, S.Ren et al introduced Faster R-CNN technology in 2015. Faster R-CNN uses a fully convolutional layer called RPN (Region Proposal Network) that runs on random images of various sizes and possibly generates a set of proposals at each feature map location. The feature vectors obtained from the derived feature map are then fed into a classification layer, followed by a bounding box regression layer for object localization, which eventually leads to object detection. Faster R-CNN can be considered as the first end-to-end and quasi-real-time deep learning detection model. Regardless of Faster R-CNN being superior to Fast R-CNN in terms of detection speed, there is still some computational inertia in successive detection stages (S. Ren, He, Girshick, & Sun, 2015; Sharma & Mir, 2020).

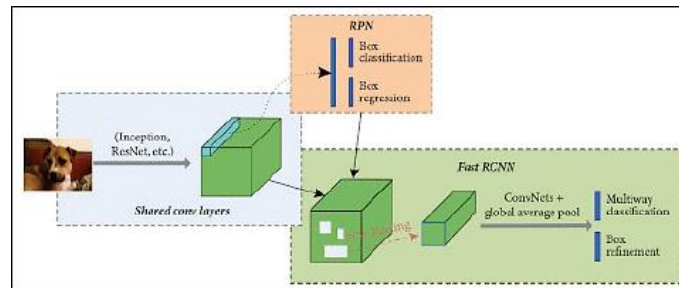


Figure 6. Architecture of Faster R-CNN(Girshick, 2015)

5) R-FCN

In 2016, Dai et al. propose a region-based fully convolutional network (R-FCN) that shares the total computational cost in the region classification step. R-FCN achieved better performance compared to Faster R-CNN without operating on region-wise fully connected layers. This is because R-FCN creates a location-aware point map that encodes comparative location information about different categories, and also uses a location-aware clustering layer to derive features of the spatially aware region by encoding each comparative location of the target region (Dai, Li, He, & Sun, 2016).

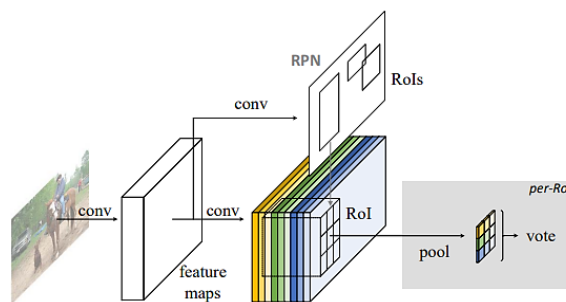


Figure 7. Overall architecture of R-FCN. A Region Proposal Network (RPN) (S. Ren et al., 2015)

6) FPN

In 2017, Lin et al proposed an FPN technique in which deep layer features are combined with shallow features to enable detection of objects at different levels. The basic idea is to develop spatially accurate features while creating good semantic features from deeper layers. Using this FPN in the traditional Faster R-CNN model provides better detection results. It should also be noted that FPN is now a fundamental building block in the latest (Lin, Dollár, et al., 2017).

7) Mask R-CNN

Mask R-CNN is a state-of-the-art Convolutional Neural Network (CNN) in terms of image segmentation and sample segmentation. It was introduced in 2017 by He et al. Segmentation as an extension of Faster R-CNN, i.e.

4) RetinaNet

First presented in 2018 by Lin et al. RetinaNet is a single object detection paradigm. The focus loss function is used to compensate for class imbalance during training. Focus loss applies a modification term to the entropy loss of focus in hard negative examples. RetinaNet is a single, unified network consisting of a core network and two task-specific subnets. The backbone is responsible for calculating the convolutional feature map across the entire input image and is an independent convolutional network. The first subnet classifies a convolutional object at the output of the backbone; The second subnet performs bounding box convolutional regression. The two subnets have a simple design that the authors recommend specifically for single-stage dense sensing (Lin, Goyal, Girshick, He, & Dollár, 2017; Zaidi et al., 2022).

Comparisons

A. Compared to traditional object detection techniques.

Table 1. Comparison of traditional object detection techniques

Techniques	Advantages	Disadvantages
SIFT	Invariance to rotation, scale changes, and affine transformations.	Low detection rate; Not good under lighting changes.
HOG	Invariance to geometric and photometric transformations.	Large amount of calculations; Low detection rate; It is difficult to manage the shelter
SURF	Fast; Achieves the same performance as SIFT.	It is not robust against rotation and lighting changes.
ORB	Faster than SURF; Excellent real-time performance	High mismatch rate; Less robust.

B. Compared to deep learning-based object detection techniques.

1) Two-stage detectors

Table 2. Comparison of traditional object detection techniques

Techniques	Advantages	Disadvantages
R-CNN	The proposed number of regions is fewer compared to the sliding window technique.	Multi-stage training; Expensive in terms of space and time; Slow object detection
SPP-Net	Faster than R-CNN; Avoids frequent computation of features.	Low resolution of the ultra-deep neural network
Fast R-CNN	High-quality detection compared to SPP-Net and R-CNN; single-stage training	Slow assembly: Selective search is slow so computation time is still high.
Faster R-CNN	Faster than Fast R-CNN.	Slow object suggestion implementation slower than YOLO.
R-FCN	Faster than R-CNN with acceptable accuracy; easier training; Reduce complexity	Need more account resources
FPN	Rich semantics at every level.	Removing top-down connections reduces accuracy
Mask R-CNN	Easy to train; Good generalization; Small overheads.	Not suitable for all real-time applications.

2) Single stage detectors

Table 3. Comparison of single stage detectors

Techniques	Advantages	Disadvantages
OverFeat	High R-CNN speed.	Less accurate.
YOLO	Efficient unified object detector; Extremely fast; Less background error than Fast R-CNN.	Unable to detect multiple objects within the same grid; Loss in accuracy rate; Possibility of detecting one object multiple times; Inability to localize small-sized objects.
SSD	Faster than Faster R-CNN; Works well for larger objects.	Does not generate sufficient high-level features for small objects.
RetinaNet	Much faster than R-CNN; Predefined anchors are not required.	Spend more time creating corners and evaluating the backbone network.

The Most Important Applications of Object Detection

Object detection applications are very widely used in many fields where it is used in the detection of remote objects such as forest fire detection and object tracking, in security and control, and in its actual uses in autonomous driving to detect unmanned aerial vehicles and other vehicles. It is also medically important as it is

used in tomography and medical imaging to detect objects. Also, in the field of agriculture etc. It is used in areas. Object detection applications have many uses that we cannot count. In this section, the most common and used applications are mentioned (Bhagya & Shyna, 2019).

A. Face recognition.

Face detection is one of the most popular object detection applications that involves detecting the presence of faces from selected photos. Convolutional neural networks play an important role in extracting and detecting facial features and reporting the final result. Social media uses this application to detect the facial image while uploading a photo and it is extremely important in the current world.

B. Vehicle detection.

Another application is car, motorcycle, bus, etc. acting as objects in this scenario. detecting the presence of vehicles. To track moving vehicles on the road, speed is a critical factor for object tracking and has proven successful.

C. Security and surveillance.

This app is highly valuable in the current world due to the increase in anti-social activities. Prevent intruders, explosives, etc. Remote sensing image-based detection falls into this category. Anomaly detection is another application that companies spend a lot of money on and it also falls into this category of object detection applications. A lot of research work is being done to automate these systems to improve performance.

Conclusion

In conclusion, object detection is a crucial aspect of computer vision, with traditional methods like SIFT, HOG, SURF, and ORB paving the way for deep learning techniques such as R-CNN, Faster R-CNN, and YOLO. While traditional methods offer invariance and speed advantages, deep learning approaches provide higher accuracy and efficiency. Moreover, object detection finds extensive applications in various fields including face recognition, vehicle detection, and security, highlighting its significance in modern technology and everyday life.

References

- Bay, H., Tuytelaars, T., & Gool, L. V. (2006). *Surf: Speeded up robust features*. Paper presented at the European conference on computer vision.
- Bhagya, C., & Shyna, A. (2019). *An overview of deep learning based object detection techniques*. Paper presented at the 2019 1st International Conference on Innovations in Information and Communication Technology (ICIICT).
- Dai, J., Li, Y., He, K., & Sun, J. (2016). R-fcn: Object detection via region-based fully convolutional networks. *Advances in neural information processing systems*, 29.
- Dalal, N., & Triggs, B. (2005). *Histograms of oriented gradients for human detection*. Paper presented at the 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05).
- Di Calafiori, D., Adzic, P., Brett, A., Cavallari, F., Chen, W., Dissertori, G., . . . Leshev, G. (2009). *The CMS ECAL Detector Control System*. Paper presented at the Proceedings of the International Conference on Accelerator and Large Experimental Physics Control Systems, Kobe, Japan.
- Diwakar, & Raj, D. (2022). Recent Object Detection Techniques: A Survey. *International Journal of Image, Graphics and Signal Processing*, 14(2), 47-60. doi:10.5815/ijigsp.2022.02.05
- Girshick, R. (2015). *Fast r-cnn*. Paper presented at the Proceedings of the IEEE international conference on computer vision.
- Girshick, R., Donahue, J., Darrell, T., & Malik, J. (2014). *Rich feature hierarchies for accurate object detection and semantic segmentation*. Paper presented at the Proceedings of the IEEE conference on computer vision and pattern recognition.
- He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). *Mask r-cnn*. Paper presented at the Proceedings of the IEEE international conference on computer vision.
- He, K., Zhang, X., Ren, S., & Sun, J. (2015). Spatial pyramid pooling in deep convolutional networks for visual recognition. *IEEE transactions on pattern analysis and machine intelligence*, 37(9), 1904-1916.
- Jha, S. K. (2022). Advances in Object Detection Algorithms–A Brief Survey.
- Kaur, R., & Singh, S. (2022). A comprehensive review of object detection with deep learning. *Digital Signal Processing*, 103812.
- Kleban, J., Xie, X., & Ma, W.-Y. (2008). *Spatial pyramid mining for logo detection in natural scenes*. Paper presented at the 2008 IEEE International Conference on Multimedia and Expo.
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6), 84-90.
- Li, K., & Cao, L. (2020). *A review of object detection techniques*. Paper presented at the 2020 5th International Conference on Electromechanical Control Technology and Transportation (ICECTT).

- Li, X., Xie, C., Jia, Y., & Zhang, G. (2013). Rapid moving object detection algorithm based on ORB features. *Journal of Electronic Measurement and Instrument*, 27(5), 455-460.
- Lin, T.-Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). *Feature pyramid networks for object detection*. Paper presented at the Proceedings of the IEEE conference on computer vision and pattern recognition.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). *Focal loss for dense object detection*. Paper presented at the Proceedings of the IEEE international conference on computer vision.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., . . . Zitnick, C. L. (2014). *Microsoft coco: Common objects in context*. Paper presented at the European conference on computer vision.
- Liu, L., Ouyang, W., Wang, X., Fieguth, P., Chen, J., Liu, X., & Pietikäinen, M. (2020). Deep learning for generic object detection: A survey. *International journal of computer vision*, 128(2), 261-318.
- Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.-Y., & Berg, A. C. (2016). *Ssd: Single shot multibox detector*. Paper presented at the European conference on computer vision.
- Lowe, D. G. (1999). *Object recognition from local scale-invariant features*. Paper presented at the Proceedings of the seventh IEEE international conference on computer vision.
- MLK. (2020). 6 Different Types of Object Detection Algorithms in Nutshell. Retrieved from <https://machinelearningknowledge.ai/different-types-of-object-detection-algorithms/>
- Ning, C., Zhou, H., Song, Y., & Tang, J. (2017). *Inception single shot multibox detector for object detection*. Paper presented at the 2017 IEEE International Conference on Multimedia & Expo Workshops (ICMEW).
- Noman, M., Stankovic, V., & Tawfik, A. (2019). *Object detection techniques: Overview and performance comparison*. Paper presented at the 2019 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT).
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). *You only look once: Unified, real-time object detection*. Paper presented at the Proceedings of the IEEE conference on computer vision and pattern recognition.
- Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems*, 28.
- Ren, Y., Zhu, C., & Xiao, S. (2018). Object detection based on fast/faster RCNN employing fully convolutional architectures. *Mathematical Problems in Engineering*, 2018, 1-7.
- Sermanet, P., Eigen, D., Zhang, X., Mathieu, M., Fergus, R., & LeCun, Y. (2013). Overfeat: Integrated recognition, localization and detection using convolutional networks. *arXiv preprint arXiv:1312.6229*.
- Sharma, V., & Mir, R. N. (2020). A comprehensive and systematic look up into deep learning based object detection techniques: A review. *Computer Science Review*, 38, 100301.
- Xiao, Y., Tian, Z., Yu, J., Zhang, Y., Liu, S., Du, S., & Lan, X. (2020). A review of object detection based on deep learning. *Multimedia Tools and Applications*, 79(33), 23729-23791.
- Zaidi, S. S. A., Ansari, M. S., Aslam, A., Kanwal, N., Asghar, M., & Lee, B. (2022). A survey of modern deep learning based object detection models. *Digital Signal Processing*, 103514.
- Zhao, Z.-Q., Zheng, P., Xu, S.-t., & Wu, X. (2019). Object detection with deep learning: A review. *IEEE transactions on neural networks and learning systems*, 30(11), 3212-3232.
- Zou, X. (2019). *A review of object detection techniques*. Paper presented at the 2019 International Conference on Smart Grid and Electrical Automation (ICSGEA).
- Zou, Z., Shi, Z., Guo, Y., & Ye, J. (2019). Object detection in 20 years: A survey. *arXiv preprint arXiv:1905.05055*.